

Security and Protection of Systems

Lecture 09 *Asymmetric Cryptography*

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Outlines

- Introduction
- RSA Algorithm
- Elgamal Algorithm
- Diffie-Hellman Algorithm

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Terminology

- **Asymmetric Keys:** Two related keys, a public(PK) and private (sk) , used to perform complementary operations, such as encryption, decryption or signature generation/verification .
- **Public key certificate:** a digital document issued and digitally signed by the sk of certification authority that binds the name of subscriber to a PK. The certificate indicates that the subscriber identified in the certificate has sole control and access the corresponding private key.
- **Public key(asymmetric) cryptographic algorithm:** a algorithm that uses two related keys. The two keys have the property that deriving sk from the PK is computationally infeasible.
- **Public key Infrastructure:** a set of policies, processes, server platforms, software and workstations used for the purpose of administration certificate sk, PK pairs, including the ability to issue, maintain, and revoke PK certificates.

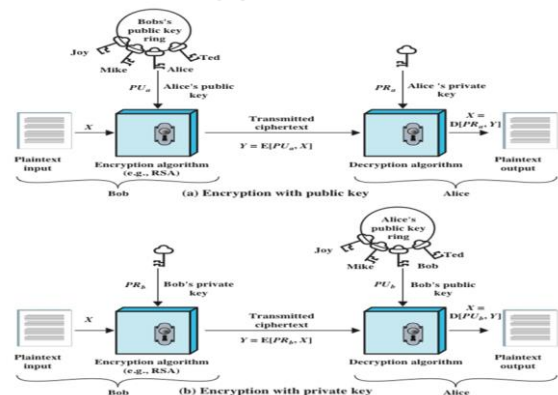
Misconceptions Concerning Public-Key Encryption

Misconceptions Concerning Public-Key Encryption

1. Public-key encryption is more secure than symmetric encryption
2. PKI is a general-purpose technique that has made symmetric encryption obsolete
3. There is a feeling that key distribution is trivial when using PKI, compared to the cumbersome handshaking involved with key distribution centers for symmetric encryption
- **Principles of Public-Key Cryptosystems** : evolved from an attempt to attack two of the most difficult problems associated with symmetric encryption:
 1. Key distribution: How to have secure communications in general without having to trust a key distribution center (KDC) with your key
 2. Digital signatures: How to verify that a message comes intact from the claimed sender

Public-Key Encryption/Decryption

1. Each user generates a pair of keys to be used for the encryption/decryption.
2. Each user places one of the two keys in a public register or accessible file (PK), the companion key is kept private (sk).
3. Each user maintains a collection of PKs obtained from others.
4. Bob sends Alice a confidential message by encrypting the message using Alice's PK.
5. Alice decrypts the message using her sk. No other recipient can decrypt the message because only Alice knows Alice's sk.
6. All participants have access to PK, and private keys are generated locally by each participant and therefore need never be distributed.



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Rivest-Shamir-Adleman (RSA) Algorithm

- by Ron Rivest, Adi Shamir, and Len Adleman at MIT and first published in 1978
- The Rivest-Shamir-Adleman (RSA) scheme has since that time reigned supreme as the most widely accepted and implemented general-purpose approach to public-key encryption.
- The RSA scheme is a cipher in which the plaintext and ciphertext are integers between 0 and $n - 1$ for some n .
- A typical size for n is 1024 bits, or 309 decimal digits.
- That is the probability of n is less than 2^{1024} .

RSA Algorithm

- Plaintext is encrypted in blocks, with each block having a binary value less than some number n .
- Block size must be less than or equal to $\log_2(n) + 1$; in practice, the block size is i bits, where $2^i < n \leq 2^{i+1}$.
- Encryption / decryption are of the following form, for some plaintext block M and ciphertext block C .

$$C = M^e \bmod n$$

$$M = C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$$

- Both sender and receiver must know the value of n . The sender knows the value of e , and only the receiver knows the value of d .
- Thus, this is a public-key encryption algorithm with a public key of $PU = \{e, n\}$ and a private key of $PR = \{d, n\}$.

Example of RSA Algorithm

Key Generation by Alice

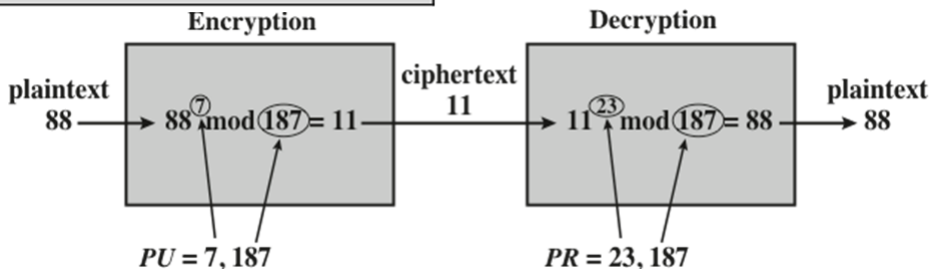
Select p, q	p and q both prime, $p \neq q$
Calculate $n = p \times q$	
Calculate $\phi(n) = (p-1)(q-1)$	
Select integer e	$\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
Calculate d	$d = e^{-1} \pmod{\phi(n)}$
Public key	$PU = \{e, n\}$
Private key	$PR = \{d, n\}$

Encryption by Bob with Alice's Public Key

Plaintext:	$M < n$
Ciphertext:	$C = M^e \bmod n$

Decryption by Alice with Alice's Private Key

Ciphertext:	C
Plaintext:	$M = C^d \bmod n$



Algorithm Requirements

- For this algorithm to be satisfactory for public-key encryption, the following requirements must be met:
 1. It is possible to find values of e, d, n such that $M^{ed} \bmod n = M$ for all $M < n$
 2. It is relatively easy to calculate $M^e \bmod n$ and $C^d \bmod n$ for all values of $M < n$
 3. It is infeasible to determine d given e and n
 - *Why? Because of Factorization Problem*

The Security of RSA

1. **Brute force:** Involves trying all possible private keys
2. **Mathematical attacks :** There are several approaches, all equivalent in effort to factoring the product of two primes
3. **Timing attacks:** These depend on the running time of the decryption algorithm
4. **Hardware fault-based attack:** This involves inducing hardware faults in the processor that is generating digital signatures
5. **Chosen ciphertext attacks:** This type of attack exploits properties of the RSA algorithm

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ElGamal Cryptography

- Announced in 1984 by T. Elgamal
- Public-key scheme based on discrete logarithms closely related to the Diffie-Hellman technique
- Used in the digital signature standard (DSS) and the S/MIME e-mail standard
- Global elements are a prime number q and a which is a primitive root of q
- Security is based on the difficulty of computing discrete logarithms

ElGamal Cryptography

Global Public Elements

q	prime number
α	$\alpha < q$ and α a primitive root of q

Key Generation by Alice

Select private X_A	$X_A < q - 1$
Calculate Y_A	$Y_A = \alpha^{X_A} \bmod q$
Public key	$\{q, \alpha, Y_A\}$
Private key	X_A

ElGamal Cryptography

Encryption by Bob with Alice's Public Key

Plaintext:	$M < q$
Select random integer k	$k < q$
Calculate K	$K = (Y_A)^k \bmod q$
Calculate C_1	$C_1 = \alpha^k \bmod q$
Calculate C_2	$C_2 = KM \bmod q$
Ciphertext:	(C_1, C_2)

Decryption by Alice with Alice's Private Key

Ciphertext:	(C_1, C_2)
Calculate K	$K = (C_1)^{X_A} \bmod q$
Plaintext:	$M = (C_2 K^{-1}) \bmod q$

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Elliptic Curve Arithmetic

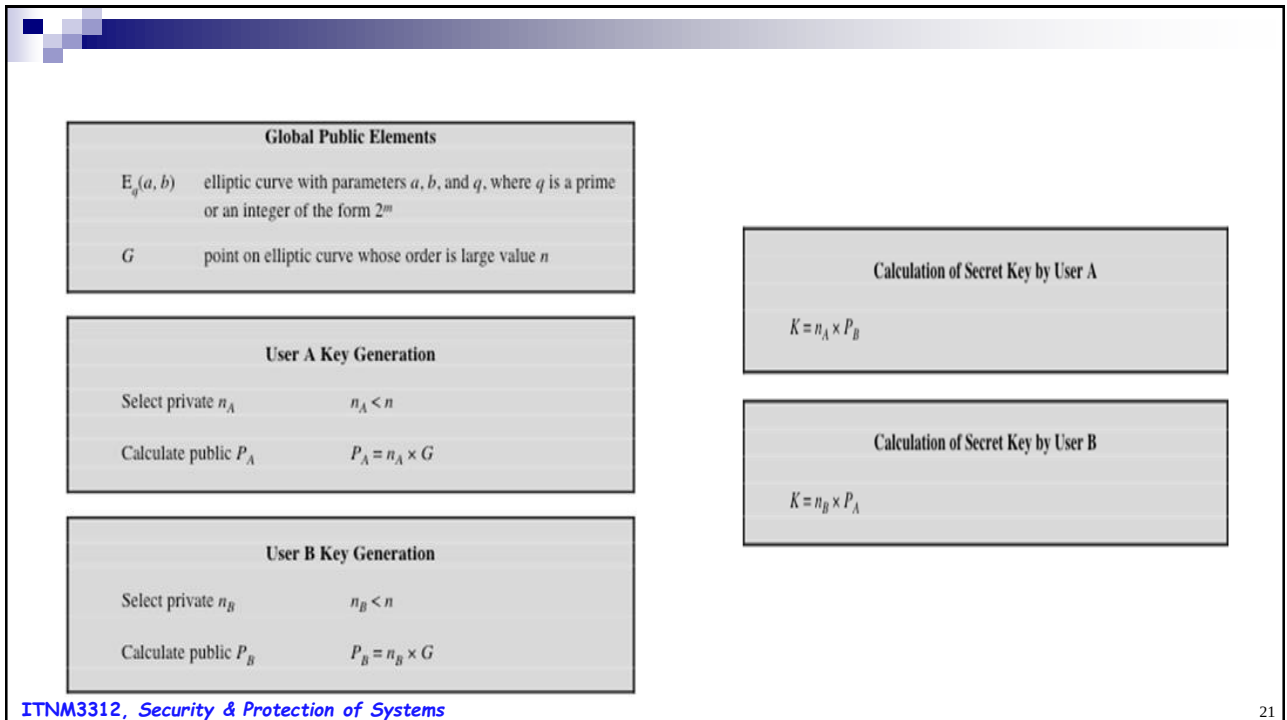
- Most of the products and standards that use public-key cryptography for encryption and digital signatures use RSA
 - The key length for secure RSA use has increased over recent years and this has put a heavier processing load on applications using RSA
- Elliptic curve cryptography (ECC) is showing up in standardization efforts including the IEEE P1363 Standard for Public-Key Cryptography
- Principal attraction of ECC is that it appears to offer equal security for a far smaller key size

Elliptic Curves Over $GF(2^m)$

- Use a cubic equation in which the variables and coefficients all take on values in $GF(2^m)$ for some number m
- Calculations are performed using the rules of arithmetic in $GF(2^m)$
- The form of cubic equation appropriate for cryptographic applications for elliptic curves is somewhat different for $GF(2^m)$ than for Z_p
 - It is understood that the variables x and y and the coefficients a and b are elements of $GF(2^m)$ and that calculations are performed in $GF(2^m)$

Elliptic Curve Cryptography (ECC)

- Addition operation in ECC is the counterpart of modular multiplication in RSA
- Multiple addition is the counterpart of modular exponentiation
- To form a cryptographic system using elliptic curves, we need to find a "hard problem" corresponding to factoring the product of two primes or taking the discrete logarithm
 - $Q = kP$, where Q, P belong to a prime curve
 - Is "easy" to compute Q given k and P
 - But "hard" to find k given Q , and P
 - Known as the elliptic curve logarithm problem



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ECC Encryption/Decryption

- Must first encode any message m as a point on the elliptic curve P_m
- Select suitable curve and point G as in Diffie-Hellman
- Each user chooses a private key n_A and generates a public key $P_A = n_A * G$
- To encrypt and send message P_m to B , A chooses a random positive integer k and produces the ciphertext C_m consisting of the pair of points:

$$C_m = \{kG, P_m + kP_B\}$$

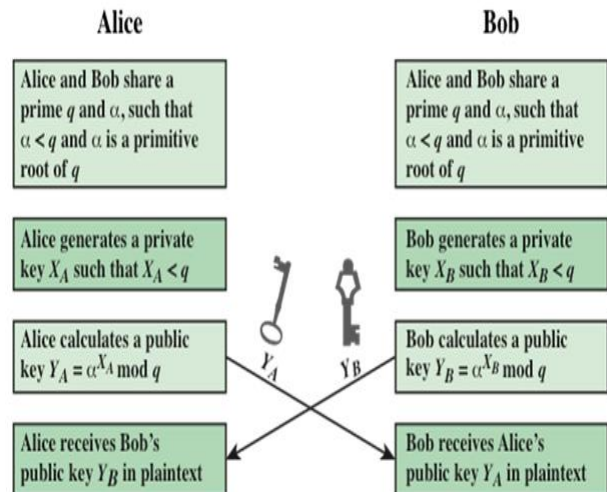
- To decrypt the ciphertext, B multiplies the first point in the pair by B 's secret key and subtracts the result from the second point:

$$P_m + kP_B - n_B(kG) = P_m + k(n_B G) - n_B(kG) = P_m$$

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Diffie-Hellman Key Exchange

- Purpose is to enable two users to securely exchange a key that can then be used for subsequent symmetric encryption of messages
- The algorithm itself is limited to the exchange of secret values
- Its effectiveness depends on the difficulty of computing discrete logarithms



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Security of Elliptic Curve Cryptography

- Depends on the difficulty of the elliptic curve logarithm problem
- Fastest known technique is "Pollard rho method"
- Compared to factoring, can use much smaller key sizes than with RSA
- For equivalent key lengths computations are roughly equivalent
- Hence, for similar security ECC offers significant computational advantages

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Symmetric and Asymmetric cryptography

Conventional Encryption	Public-Key Encryption
<i>Needed to Work:</i> 1. The same algorithm with the same key is used for encryption and decryption. 2. The sender and receiver must share the algorithm and the key. <i>Needed for Security:</i> 1. The key must be kept secret. 2. It must be impossible or at least impractical to decipher a message if the key is kept secret. 3. Knowledge of the algorithm plus samples of ciphertext must be insufficient to determine the key.	<i>Needed to Work:</i> 1. One algorithm is used for encryption and a related algorithm for decryption with a pair of keys, one for encryption and one for decryption. 2. The sender and receiver must each have one of the matched pair of keys (not the same one). <i>Needed for Security:</i> 1. One of the two keys must be kept secret. 2. It must be impossible or at least impractical to decipher a message if one of the keys is kept secret. 3. Knowledge of the algorithm plus one of the keys plus samples of ciphertext must be insufficient to determine the other key.

Comparable Key Sizes in Terms of Computational Effort for Cryptanalysis

Symmetric key algorithms	Diffie-Hellman, Digital Signature Algorithm	RSA (size of n in bits)	ECC (modulus size in bits)
80	$L = 1024$ $N = 160$	1024	160–223
112	$L = 2048$ $N = 224$	2048	224–255
128	$L = 3072$ $N = 256$	3072	256–383
192	$L = 7680$ $N = 384$	7680	384–511
256	$L = 15,360$ $N = 512$	15,360	512+

L = size of PK
 N = size of sk

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie-Hellman	No	No	Yes
DSS	No	Yes	No

Applications
for PKI



Thanks